Scientific Computing 2026

# Joint distributions, dependence, and random vectors

Lecture 3

# Two responses from the same user

One user has click probability $P = 0.1$ or $0.9$, equally likely.
Given $P$, the responses $X, Y \in \{0, 1\}$ are independent.

**Definition: Joint probability mass function**

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$$

| | $Y = 0$ | $Y = 1$ | Total |
|---|---|---|---|
| $X = 0$ | $0.41$ | $0.09$ | $0.50$ |
| $X = 1$ | $0.09$ | $0.41$ | $0.50$ |
| Total | $0.50$ | $0.50$ | $1$ |

$$p_{X,Y}(1, 1) = \tfrac{1}{2}(0.1)^2 + \tfrac{1}{2}(0.9)^2 = 0.41.$$

# Marginals discard the pairing

## Definition: Marginal PMFs

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_Y(y) = \sum_x p_{X,Y}(x,y)$$

**Same user**

$$(p_{X,Y}(x,y)) = \begin{pmatrix} 0.41 & 0.09 \\ 0.09 & 0.41 \end{pmatrix}$$

Agreement is more likely.

**Independently selected users**

$$(p_{X,Y}(x,y)) = \begin{pmatrix} 0.25 & 0.25 \\ 0.25 & 0.25 \end{pmatrix}$$

All pairs are equally likely.

Both models have $X, Y \sim \mathrm{Bernoulli}(0.5)$. The marginals do not determine the joint distribution.

# A conditional PMF normalizes one row

For the same user,

$$\mathbb{P}(Y = 1 \mid X = 1) = \frac{0.41}{0.50} = 0.82, \qquad \mathbb{P}(Y = 1 \mid X = 0) = \frac{0.09}{0.50} = 0.18.$$

## Definition: Conditional PMF

For $p_X(x) > 0$,

$$p_{Y|X}(y \mid x) = \frac{p_{X,Y}(x, y)}{p_X(x)}.$$

Rearranging gives the factorization

$$p_{X,Y}(x, y) = p_X(x)\, p_{Y|X}(y \mid x).$$

# Independence depends on what we condition on

Independence requires

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \qquad \text{for every } x, y.$$

The shared-user table fails: $0.41 \neq 0.5 \cdot 0.5$.

**Definition: Conditional independence given $P$**

Within each user type, the joint PMF factors:

$$p_{X,Y|P}(x, y \mid p) = p_{X|P}(x \mid p)\, p_{Y|P}(y \mid p).$$

The first click provides information about the user's probability $P$. This does not mean that one click causes the other.

# Joint densities assign probability to regions

Near $(x, y)$, a small rectangle has probability approximately

$$f_{X,Y}(x, y)\,\Delta x\,\Delta y.$$

## Definition: Joint density

A nonnegative function with total integral one, satisfying

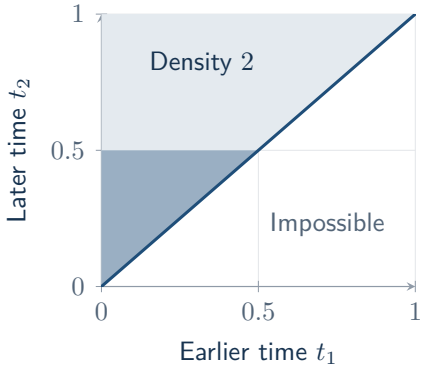
$$\mathbb{P}((X, Y) \in R) = \iint_R f_{X,Y}(x, y)\,dx\,dy$$

for regions $R$ in the plane.

Integrating over a region adds the probabilities of its small rectangles.

# Two arrivals, recorded in order

Independent $U, V \sim \mathrm{Uniform}(0,1)$, in hours.
Record $T_1 = \min(U,V)$ and $T_2 = \max(U,V)$.




Either person can arrive first:

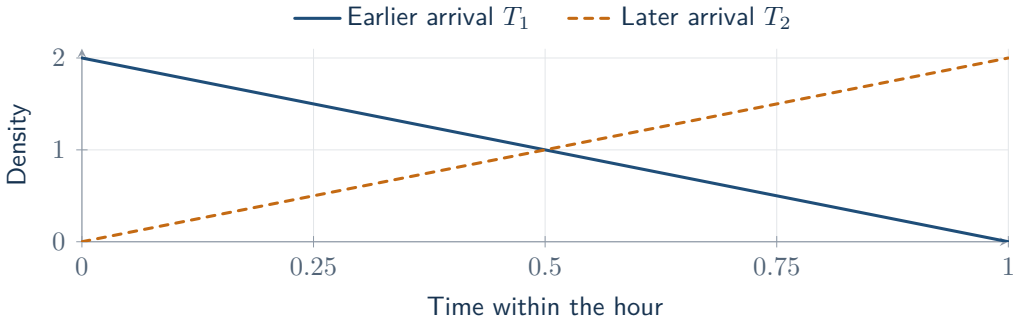
$$f_{T_1,T_2}(t,s) = 2, \quad 0 < t < s < 1.$$

Density is zero elsewhere. Both arrive before half an hour:

$$\mathbb{P}(T_2 \le \tfrac{1}{2}) = 2 \cdot \frac{1}{8} = \frac{1}{4}.$$

Sorting introduces dependence: the later arrival must follow the earlier one.

# Marginals integrate over the other arrival




$$f_{T_1}(t) = \int_t^1 2\,ds = 2(1-t), \qquad f_{T_2}(s) = \int_0^s 2\,dt = 2s.$$

An early first arrival leaves more possible times for the second. In general, $f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x,y)\,dy$.

# A conditional density normalizes one slice

Given $T_1 = t$, the later arrival can be anywhere in $(t, 1)$.

**Definition: Conditional density**

Where $f_X(x) > 0$,

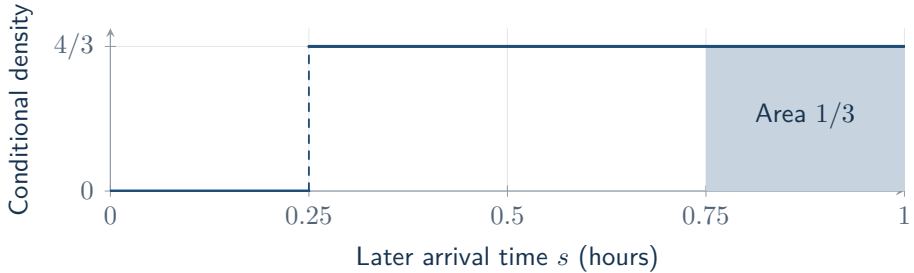
$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

$$f_{T_2|T_1}(s \mid t) = \frac{2}{2(1-t)} = \frac{1}{1-t}, \quad t < s < 1.$$

For smooth densities, this is the limit of conditioning on a small interval around $x$. The denominator is a density.

# Conditional probability of a late second arrival

The first arrival is at $t = \frac{1}{4}$. The second is uniform over the remaining $\frac{3}{4}$ hour.




$$\mathbb{P}(T_2 > \tfrac{3}{4} \mid T_1 = \tfrac{1}{4}) = \int_{3/4}^{1} \frac{4}{3}\, ds = \frac{1}{3}.$$

The shaded area is the probability that the second arrival occurs in the final quarter-hour, given the first arrival at 15 minutes.

# The expected gap between arrivals

The expectation rule now uses the joint density:

$$\mathbb{E}[g(X,Y)] = \iint_{\mathbb{R}^2} g(x,y) f_{X,Y}(x,y)\,dx\,dy.$$

For the gap $g(t,s) = s - t$,

$$\begin{aligned}\mathbb{E}[T_2 - T_1] &= \int_0^1 \int_t^1 (s-t)\,2\,ds\,dt \\ &= \int_0^1 (1-t)^2\,dt = \frac{1}{3}\ \text{hour}.\end{aligned}$$

For a discrete pair, replace the integrals by sums. A finite expectation requires $\mathbb{E}[|g(X,Y)|] < \infty$.

# Correlation removes the units

For the click table,

$$\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.41 - 0.25 = 0.16.$$

## Definition: Correlation

For finite, positive standard deviations $\sigma_X, \sigma_Y$,

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y} = \mathbb{E}\left[\frac{X - \mu_X}{\sigma_X}\frac{Y - \mu_Y}{\sigma_Y}\right].$$

For the clicks, $\sigma_X = \sigma_Y = 0.5$, so $\rho_{X,Y} = 0.64$.

Changing minutes to seconds changes covariance but leaves correlation unchanged.

# Why correlation lies between $-1$ and $1$

Standardize: $U = (X - \mu_X)/\sigma_X$, $V = (Y - \mu_Y)/\sigma_Y$.

$$0 \le \mathbb{E}[(U - V)^2] = 2 - 2\rho, \qquad 0 \le \mathbb{E}[(U + V)^2] = 2 + 2\rho.$$

**Key result: Correlation bound and equality cases**

For finite, positive marginal variances, $-1 \le \rho \le 1$.

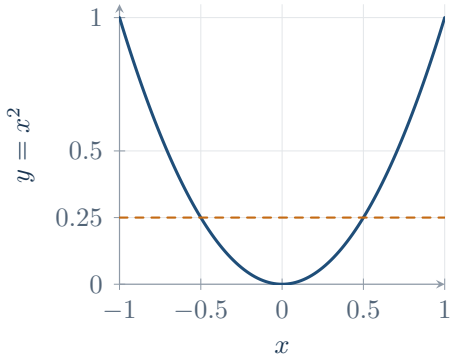
$\rho = 1 \iff U = V$ almost surely, $\qquad \rho = -1 \iff U = -V$ almost surely.

Perfect correlation means an exact affine relationship.

If either variable is constant, correlation is undefined.

# Zero correlation can hide complete dependence

Let $X \sim \mathrm{Uniform}(-1, 1)$ and $Y = X^2$.




Symmetry gives

$$\mathbb{E}[X] = \mathbb{E}[X^3] = 0.$$

Therefore

$$\mathrm{Cov}(X, Y) = 0, \qquad \rho_{X,Y} = 0.$$

Yet knowing $X$ determines $Y$ exactly.

$\mathbb{P}(Y \le 1/4 \mid |X| \le 1/2) = 1$, while $\mathbb{P}(Y \le 1/4) = 1/2$. Independence fails.

# Conditional means minimize squared prediction error

Use $X$ to predict $Y$. Write

$$m(x) = \mathbb{E}[Y \mid X = x], \qquad v(x) = \mathrm{Var}(Y \mid X = x).$$

**Theorem: Best prediction under squared error**

If $\mathbb{E}[Y^2] < \infty$, every rule $a(X)$ with finite mean squared error satisfies

$$\mathbb{E}[(Y - a(X))^2] = \mathbb{E}[v(X)] + \mathbb{E}[(m(X) - a(X))^2].$$

The minimum is $\mathbb{E}[v(X)]$, attained by $a(X) = m(X)$.

The two terms are remaining uncertainty and the extra error from using a different prediction.

# Why the conditional mean is optimal

Inside the group $X = x$, split the prediction error:

$$Y - a(x) = (Y - m(x)) + (m(x) - a(x)).$$

The cross term averages to zero because

$$\mathbb{E}[Y - m(x) \mid X = x] = 0.$$

Squaring and taking conditional expectation gives

$$\mathbb{E}[(Y - a(x))^2 \mid X = x] = v(x) + (m(x) - a(x))^2.$$

Total expectation averages this identity over $X$.

# Prediction in the click model

Without the first response, the best constant prediction is $0.5$.

$$\mathbb{E}[(Y - 0.5)^2] = 0.25.$$

After observing it,

$$m(0) = 0.18, \qquad m(1) = 0.82.$$

Both groups have the same conditional variance:

$$\mathbb{E}[(Y - m(X))^2] = 0.18(0.82) = 0.1476.$$

# Several measurements on one object

Record several sensor readings, or several features of one observation.

**Definition: Random vector and mean vector**

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_d \end{pmatrix} \in \mathbb{R}^d, \qquad \boldsymbol{\mu} = \mathbb{E}[\mathbf{X}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_d] \end{pmatrix}$$

All coordinates belong to the same object or experimental repetition.
A realized observation is a lowercase vector $\mathbf{x}$.

The joint distribution assigns probabilities to regions in $\mathbb{R}^d$.

# A covariance matrix records every pair

## Definition: Covariance matrix

For a random vector with finite second moments,

$$\Sigma = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^{\top}], \qquad \Sigma_{ij} = \operatorname{Cov}(X_i, X_j).$$

Diagonal entries are variances. Off-diagonal entries are covariances.

For the two clicks,

$$\boldsymbol{\mu} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 0.25 & 0.16 \\ 0.16 & 0.25 \end{pmatrix}.$$

Standardizing the coordinates gives $R_{ij} = \Sigma_{ij} / \sqrt{\Sigma_{ii}\Sigma_{jj}}$. Here $R$ has diagonal $1$ and off-diagonal $0.64$.

# Every weighted sum must have nonnegative variance

For $S = \mathbf{a}^\top \mathbf{X} = \sum_i a_i X_i$,

$$\operatorname{Var}(S) = \sum_i \sum_j a_i a_j \Sigma_{ij} = \mathbf{a}^\top \Sigma \mathbf{a} \geq 0.$$

**Definition: Positive semidefinite matrix**

A symmetric matrix $\Sigma$ is positive semidefinite when

$$\mathbf{a}^\top \Sigma \mathbf{a} \geq 0 \qquad \text{for every real } \mathbf{a}.$$

It is positive definite when this is strictly positive for every nonzero $\mathbf{a}$.

Every covariance matrix is positive semidefinite. A zero-variance combination can make it singular.

# Positive diagonal entries are not enough

Could this be a covariance matrix?

$$M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

It would imply

$$\mathrm{Var}(X_1 - X_2) = 1 + 1 - 2(2) = -2.$$

**Key result: Valid covariance matrices in two dimensions**

$$\begin{pmatrix} v_1 & c \\ c & v_2 \end{pmatrix} \succeq 0 \quad \Longleftrightarrow \quad v_1 \geq 0, \quad v_2 \geq 0, \quad c^2 \leq v_1 v_2$$

With positive variances, the last condition is exactly $|\rho| \leq 1$. A diagonal covariance matrix alone does not guarantee independence.

# Means and covariances after a linear transformation

Let $\mathbf{W} = A\mathbf{X} + \mathbf{b}$, with fixed $A$ and $\mathbf{b}$.

**Theorem: Linear transformation of moments**

For $\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}$ and finite covariance $\Sigma$,

$$\mathbb{E}[\mathbf{W}] = A\boldsymbol{\mu} + \mathbf{b}, \qquad \mathrm{Cov}(\mathbf{W}) = A\Sigma A^{\top}.$$

**Proof.** Centering removes the shift:

$$\mathbf{W} - \mathbb{E}[\mathbf{W}] = A(\mathbf{X} - \boldsymbol{\mu}).$$

Take the expected outer product and pull out the fixed matrices.

These rules need neither independence nor a Gaussian distribution.

# Average error and disagreement between sensors

Centered errors have $\Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}$.

$$M = \frac{X_1 + X_2}{2}, \quad D = X_1 - X_2, \quad A = \begin{pmatrix} 1/2 & 1/2 \\ 1 & -1 \end{pmatrix}.$$

$$\operatorname{Cov}\begin{pmatrix} M \\ D \end{pmatrix} = A\Sigma A^\top = \begin{pmatrix} 0.9 & 0 \\ 0 & 0.4 \end{pmatrix}.$$

Averaging retains much of the shared error.
Taking the difference cancels much of it.

With independent unit-variance errors, the variances would be $0.5$ and $2$. Zero covariance between $M$ and $D$ does not yet establish independence.

# Correlated Gaussian readings from independent noise

Take independent $Z_1, Z_2 \sim \mathcal{N}(0,1)$ and set

$$X_1 = Z_1, \qquad X_2 = 0.8Z_1 + 0.6Z_2.$$

Both sensors contain the disturbance $Z_1$.
Only the second contains the additional disturbance $Z_2$.

$$\mathrm{Var}(X_2) = 0.8^2 + 0.6^2 = 1, \qquad \mathrm{Cov}(X_1, X_2) = 0.8.$$

Both marginals are standard Gaussian.

# The multivariate Gaussian construction

**Definition: Multivariate Gaussian**

Let $\mathbf{Z}$ have independent standard Gaussian coordinates. For fixed $A$ and $\boldsymbol{\mu}$,

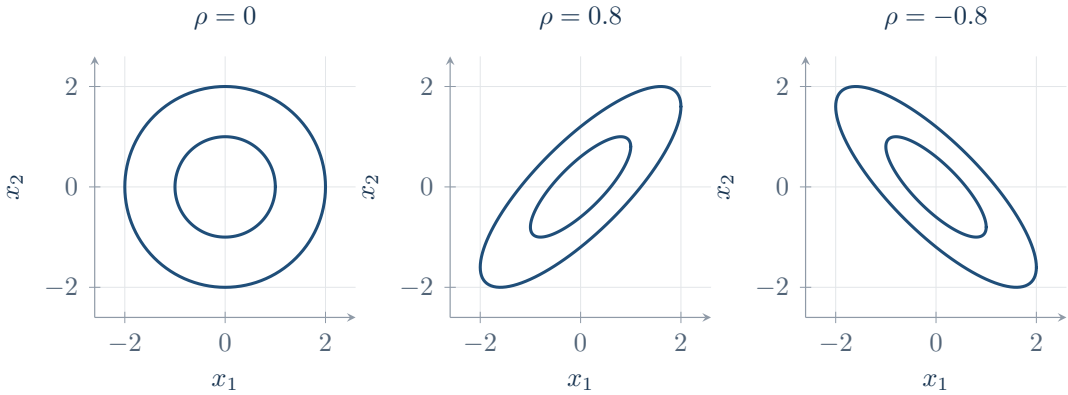
$$\mathbf{X} = \boldsymbol{\mu} + A\mathbf{Z} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma), \qquad \Sigma = AA^\top.$$

Every linear combination is Gaussian:

$$\mathbf{a}^\top \mathbf{X} \sim \mathcal{N}(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \Sigma \mathbf{a}).$$

Zero variance means a constant. The rows of $A$ specify which independent noises each measurement shares.

# Correlation changes the joint shape




In every panel, both marginals are $\mathcal{N}(0, 1)$. Each curve joins points of equal density. Outer curves have lower density.

# The bivariate density from a change of coordinates

For $|\rho| < 1$, write

$$X_1 = Z_1, \qquad X_2 = \rho Z_1 + \sqrt{1-\rho^2}\, Z_2.$$

The inverse uses $z_1 = x_1$, $z_2 = (x_2 - \rho x_1)/\sqrt{1-\rho^2}$. The transformation multiplies areas by $\sqrt{1-\rho^2}$.

**Key result: Standard bivariate Gaussian density**

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1-\rho^2)}\right]$$

Substitute the inverse into the independent-noise density and divide by the area scale to preserve probability.

# The long and short directions of a Gaussian ellipse

For $\rho = 0.8$, use rotated coordinates

$$U = \frac{X_1 + X_2}{\sqrt{2}}, \qquad V = \frac{X_1 - X_2}{\sqrt{2}}.$$

Their variances are $1.8$ and $0.2$, with covariance zero.

The exponent depends on

$$q = \frac{u^2}{1.8} + \frac{v^2}{0.2}.$$

At $q = c$, the axis lengths are proportional to $\sqrt{1.8c}$ and $\sqrt{0.2c}$.

The equal-sign direction is three times longer. Shared sensor fluctuations stretch the distribution along the diagonal.

# The Gaussian density in matrix notation

**Key result: Multivariate Gaussian density**

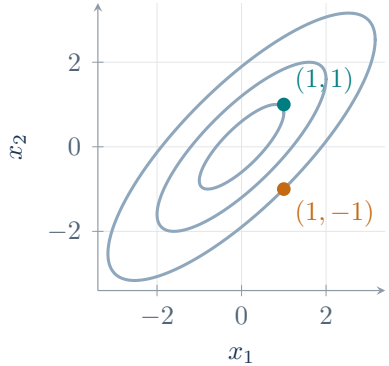
For a positive-definite covariance matrix $\Sigma$,

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{\exp[-\frac{1}{2}q(\mathbf{x})]}{(2\pi)^{d/2}\sqrt{\det \Sigma}}, \qquad q(\mathbf{x}) = (\mathbf{x}-\boldsymbol{\mu})^{\top}\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu}).$$

$q(\mathbf{x})$ is the sum of squared independent-noise coordinates.
$\sqrt{\det \Sigma}$ accounts for the change in volume.

$\Sigma^{-1}$ is the inverse and $\det \Sigma$ the determinant. For diagonal $\Sigma$, the density factors into univariate Gaussian densities.

# Equal ordinary distance, different Gaussian density




For unit variances and $\rho = 0.8$,

$$q(1,1) = \frac{10}{9}, \qquad q(1,-1) = 10.$$

Both points are distance $\sqrt{2}$ from the origin. The disagreement $(1, -1)$ has much lower density.

**Definition: Squared Mahalanobis distance**

$$q(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

# Gaussian transformations and independence

**Theorem: Closure under linear transformations**

If $\mathbf{X} \sim \mathcal{N}_d(\boldsymbol{\mu}, \Sigma)$, then for fixed $B, \mathbf{b}$,

$$B\mathbf{X} + \mathbf{b} \sim \mathcal{N}(B\boldsymbol{\mu} + \mathbf{b},\ B\Sigma B^\top).$$

Selecting coordinates gives Gaussian marginals.

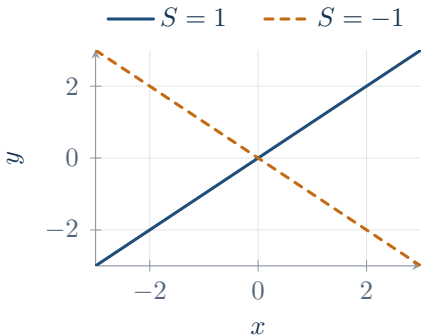
**Theorem: The joint Gaussian independence exception**

For jointly Gaussian variables, zero covariance implies independence.

Thus the sensor average and difference are independent under the joint Gaussian model. The moment rules now specify their full distribution.

# Gaussian marginals do not guarantee a joint Gaussian

Take $X \sim \mathcal{N}(0,1)$ and an independent random sign $S = \pm 1$, each with probability $1/2$. Set $Y = SX$.




$Y \sim \mathcal{N}(0,1)$, and

$$\operatorname{Cov}(X, Y) = \mathbb{E}[SX^2] = 0.$$

But $|Y| = |X|$ always.
The variables are dependent. $X + Y$ equals zero with probability $1/2$, but is not constant.

A joint Gaussian would make every linear combination Gaussian. This one has a point mass, so the pair is not jointly Gaussian.

# A singular Gaussian lives on a lower-dimensional set

Let $X_2 = X_1 \sim \mathcal{N}(0, 1)$.

$$\Sigma = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \det \Sigma = 0, \qquad \operatorname{Var}(X_1 - X_2) = 0.$$

Every observation lies on the line $x_2 = x_1$.

This is a valid Gaussian random vector, but it has no density with respect to two-dimensional area.

The Gaussian construction allows singular covariance matrices. The full-dimensional density formula requires positive definiteness.

# Observing one sensor reveals the shared noise

Recall the construction

$$X_1 = Z_1, \qquad X_2 = 0.8Z_1 + 0.6Z_2.$$

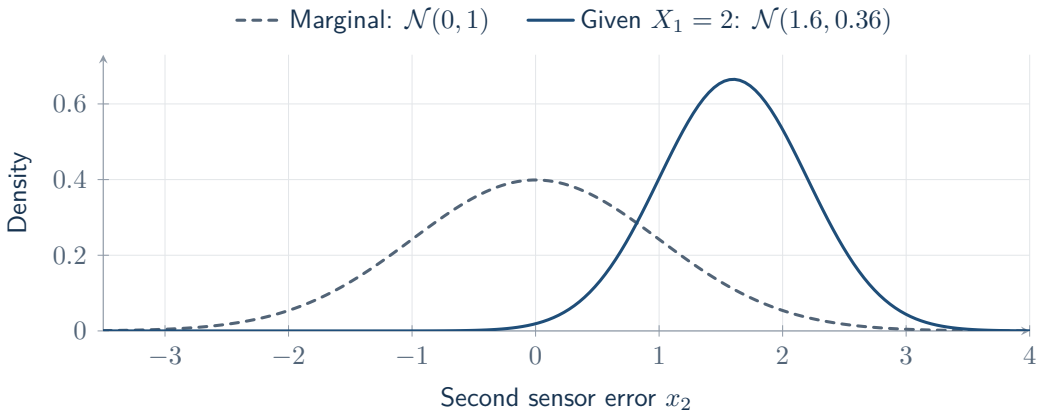
Observing $X_1 = x$ fixes $Z_1 = x$. Only $Z_2$ remains random.

$$X_2 \mid X_1 = x \sim \mathcal{N}(0.8x, 0.36).$$

If the first sensor reads two units too high, the second has conditional mean $1.6$ and conditional standard deviation $0.6$.

The marginal distribution of $X_2$ remains $\mathcal{N}(0, 1)$.

# Conditioning shifts the mean and reduces the spread




At $X_1 = 2$, the second sensor's mean shifts from $0$ to $1.6$. Its variance falls from $1$ to $0.36$.

# The general conditional Gaussian formula

## Theorem: Conditioning a bivariate Gaussian

For jointly Gaussian $X, Y$ with $\sigma_X, \sigma_Y > 0$ and $|\rho| < 1$,

$$Y \mid X = x \sim \mathcal{N}\left(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X),\ \sigma_Y^2(1 - \rho^2)\right).$$

The mean's slope is $\mathrm{Cov}(X, Y)/\mathrm{Var}(X)$.
The conditional variance is constant in $x$ in this model.

For the sensors, total variance gives

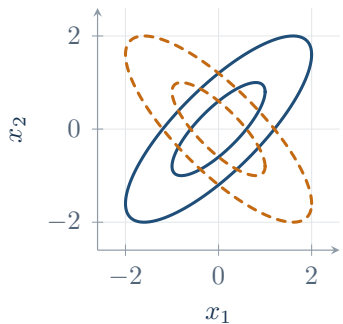
$$\underbrace{1}_{\text{total variance}} = \underbrace{0.36}_{\text{residual variance}} + \underbrace{0.64}_{\text{variance of conditional mean}}.$$

At $|\rho| = 1$, the relationship is exact and the conditional distribution is a point mass.

# Two operating modes with identical feature marginals

Two equally likely modes, both with mean zero:

$$\Sigma_+ = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}, \qquad \Sigma_- = \begin{pmatrix} 1 & -0.8 \\ -0.8 & 1 \end{pmatrix}.$$




Mode $+$: errors move together.

Mode $-$: errors move in opposite directions. Each feature alone is $\mathcal{N}(0, 1)$ in either class.

# Agreement between features identifies the mode

The determinants are equal, so the density normalization cancels:

$$\log \frac{f_+(x_1, x_2)}{f_-(x_1, x_2)} = \frac{2(0.8)}{1 - 0.8^2}\, x_1 x_2 = \frac{40}{9} x_1 x_2.$$

With equal priors, choose $+$ if $x_1 x_2 > 0$, and $-$ if $x_1 x_2 < 0$.

At $(1, 1)$,

$$\mathbb{P}(C = + \mid \mathbf{X} = (1, 1)) = \frac{1}{1 + e^{-40/9}} \approx 0.9884.$$

At $(1, -1)$, the probability is approximately $0.0116$.

Along either coordinate axis the posterior probabilities tie.

# An independence assumption can erase the signal

Suppose we replace each class's joint density by the product of its feature marginals.

$$\widetilde{f}_+(x_1, x_2) = \phi(x_1)\phi(x_2) = \widetilde{f}_-(x_1, x_2),$$

where $\phi$ is the standard Gaussian density.

Both approximate models now assign the same density to every pair.

$$\widetilde{\mathbb{P}}(C = + \mid \mathbf{X} = \mathbf{x}) = \frac{1}{2}.$$

# The Gaussian classification score

Model $\mathbb{P}(C = k) = \pi_k$ and $\mathbf{X} \mid C = k \sim \mathcal{N}_d(\boldsymbol{\mu}_k, \Sigma_k)$.

**Key result: Posterior-maximizing class score**

For $\pi_k > 0$ and positive-definite $\Sigma_k$, choose the largest

$$s_k(\mathbf{x}) = \log \pi_k - \frac{1}{2} \log \det \Sigma_k - \frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^\top \Sigma_k^{-1}(\mathbf{x} - \boldsymbol{\mu}_k).$$

The terms account for class prevalence, density spread, and displacement relative to the class covariance.

The omitted $-\frac{d}{2}\log(2\pi)$ is identical for every class.

# Shared covariance gives linear decision boundaries

If every class has the same covariance $\Sigma$, the quadratic term $-\frac{1}{2}\mathbf{x}^\top\Sigma^{-1}\mathbf{x}$ cancels in comparisons.

The class-dependent score becomes

$$\mathbf{x}^\top\Sigma^{-1}\boldsymbol{\mu}_k - \frac{1}{2}\boldsymbol{\mu}_k^\top\Sigma^{-1}\boldsymbol{\mu}_k + \log\pi_k.$$

This is linear in the observed features, plus a constant.

With different covariance matrices, quadratic terms generally remain. The two-mode example uses the interaction $x_1x_2$.

# What the joint model lets us compute

| Operation | What it answers |
|---|---|
| Marginalize | What remains if we ignore a measurement? |
| Condition | What changes after we observe a measurement? |
| Transform | How do combined measurements vary? |
| Compare class densities | Which class best explains the whole observation? |

Dependence can matter even when each individual feature looks identical across models.